MULTI CONTAINER RUNTIME

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System Architecture

engine.c

User-Space Supervisor

Container lifecycle · CLI commands · Logging pipeline · Signal handling · select() event loop

monitor.ko

Kernel Module

RSS enforcement · Timer callback · softirq context · ioctl interface · spinlock protection

Bounded Buffer

Producer-Consumer Queue

Decouples pipe reads from disk I/O · mutex + cond var · Backpressure via blocking producers

Namespaces

PID + UTS + Mount

Per-container process tree · private hostname · isolated mount table · clone() with flags

All four systems interlock: namespaces isolate, pipes stream, buffer absorbs, kernel enforces.

Linux Namespaces & Isolation

What is a Namespace?

A namespace is a kernel abstraction that makes a process believe it has an exclusive, private view of a system resource.

Without namespaces every process on Linux shares a single flat global view.

This runtime uses three: PID, UTS, and Mount — sufficient to demonstrate meaningful isolation.

Namespace	Clone Flag	What It Isolates
PID	CLONE_NEWPID	Process IDs. Container starts at PID 1
UTS	CLONE_NEWUTS	Hostname and NIS domain name
Mount	CLONE_NEWNS	Filesystem mount table. chroot becomes safe

PID Mapping Example

HOST VIEW	CONTAINER VIEW
PID 1 = systemd	PID 1 = /bin/sh ← same kernel PID
PID 4521 = engine	PID 2 = cpu_hog
PID 4522 = sh (alpha)	(no visibility above PID 1)

clone() — create container process

```
pid_t pid = clone(
  child_fn,        // container entry point
  stack_top,       // x86 stacks grow DOWN
  CLONE_NEWPID |
  CLONE_NEWUTS |
  CLONE_NEWNS  |
  SIGCHLD,         // enables waitpid()
  &cfg
);
```

Pipes & Inter-Process Communication

CHANNEL A — Pipe (log data)

container stdout/stderr → pipe buffer (64 KiB FIFO) → supervisor reader thread

- pipe2(pipefd, O_CLOEXEC)
- dup2(pipefd[1], STDOUT_FILENO) ← redirect stdout to pipe
- close(pipefd[1]) in parent ← enables EOF detection

CHANNEL B — UNIX Socket (CLI control)

/tmp/mini_runtime.sock

- engine start / stop / ps / logs
- Fixed-size packed structs — no framing
- __attribute__((packed)) prevents padding mismatches
- Kernel passes peer PID/UID via SO_PEERCRED

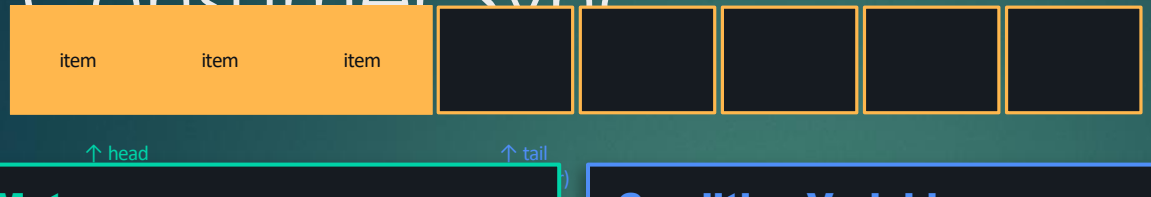
⚠ Critical: close(pipefd[1]) in parent — EOF on a pipe occurs ONLY when ALL write ends are closed. If the parent holds the write end open, the reader thread blocks forever waiting for data that never ends.

Wire Protocol — packed structs over socket

```
typedef struct { cmd_kind_t kind; char id[ID_LEN]; char rootfs[PATH_MAX]; char command[256];
  unsigned long soft_limit_bytes; unsigned long hard_limit_bytes; int nice_value;
} __attribute__((packed)) ctrl_request_t;
```

Bounded Buffer — Producer-Consumer Sync

Ring Buffer



Mutex

Binary lock — only one thread modifies head/tail/count at a time. pthread_mutex_lock/unlock.

Condition Variable

Waiting point tied to a predicate. pthread_cond_wait() atomically releases mutex AND sleeps — eliminates lost wakeup.

while() not if()

POSIX permits spurious wakeups. if() lets a woken thread proceed on a false alarm — causes overflow. while() rechecks every time.

The Lost Wakeup Bug — Why It Cannot Happen Here

Thread A checks count == 64 (full) → about to sleep. Thread B decrements count to 63 → signals not_full. Thread A sleeps forever (missed signal). pthread_cond_wait() prevents this: it holds the mutex during predicate check and atomically releases + sleeps. B cannot signal between check and sleep — it must acquire the same mutex.

```
bb_push: pthread_mutex_lock → while(count==CAP) cond_wait(&not_full, &mutex) → insert → count++ → signal(&not_empty) → unlock
```

Kernel Module: monitor.ko

User Space (ring 3): engine.c — protected; can crash safely | Kernel Space (ring 0): monitor.c — no protection; one bad pointer = kernel panic

ioctl — Crossing the Kernel Boundary

```
/* User space */
ioctl(monitor_fd, MONITOR_REGISTER, &req);

/* Kernel — monitor_ioctl() */
struct monitor_register_req req;
if (copy_from_user(&req,
     (void __user *)arg,
     sizeof(req)))
  return -EFAULT;
// NEVER dereference arg directly!
```

Why Enforce in Kernel Space?

1. CPU starvation immunity
Timer fires regardless of user-space scheduling.

2. Compromise resistance
Container cannot ptrace() or kill a kernel timer.

3. Authoritative data
Reads directly from mm_struct — kernel ground truth.

4. Unstoppable SIGKILL
force=1 bypasses signal blocking; delivered next timeslice.

Soft vs Hard Limits

Limit	Threshold	Action
Soft	soft_limit_bytes	Log kernel warning (once per container)
Hard	hard_limit_bytes	send_sig(SIGKILL, task, 1) — force=1 bypasses blocking

Signal Handling

Signal	Used For
SIGCHLD	Reap dead container processes via waitpid()
SIGTERM / SIGINT	Graceful supervisor shutdown — drain logs, stop containers
SIGPIPE	Ignored (SIG_IGN) — container closed pipe; write() returns EPIPE
SIGKILL	Sent by kernel monitor on hard limit breach (uncatchable)

Async-Signal-Safety Rule

Signal handlers can interrupt ANY point — inside malloc, printf, pthread_mutex_lock.

NEVER call malloc/free, printf, or FILE* I/O inside a handler.

Use the flag pattern: handler sets volatile sig_atomic_t; real work runs in the event loop.

```
volatile sig_atomic_t g_got_sigchld = 0;

static void sigchld_handler(int sig) {
  g_got_sigchld = 1; // ONLY safe op
}

// In event loop:
if (g_got_sigchld) {
  g_got_sigchld = 0;
  reap_children();
}
```

Reaping Children — waitpid Loop

while ((pid = waitpid(-1, &status, WNOHANG)) > 0) { /* classify exit */ } ← Loop required: Linux may coalesce multiple SIGCHLD signals into one delivery. Without the loop, unreaped zombies accumulate until the PID table exhausts.

Main Thread

select() event loop

Waits on control socket with 1s timeout. Accepts CLI connections, dispatches handlers. Checks SIGCHLD flag. Handles shutdown.

Logger Thread

shared, single instance

bb_pop() from bounded buffer. write() log lines to <id>.log on disk. On shutdown: drain all items first.

Reader Thread

one per container

read() from container's pipe write end. bb_push() into bounded buffer. Exits when pipe returns EOF.

Container Launch Sequence (CMD_START)

1. Validate: id unused, slot available
2. Open log file: logs/<id>.log
3. pipe2(pipefd, O_CLOEXEC)
4. malloc(STACK_SIZE) — clone stack
5. `clone(child_fn, …CLONE_NEWPID|UTS|NS|SIGCHLD…)`
6. free(stack); close(pipefd[1]) in parent
7. `ioctl MONITOR_REGISTER — kernel module`
8. pthread_create — start reader thread

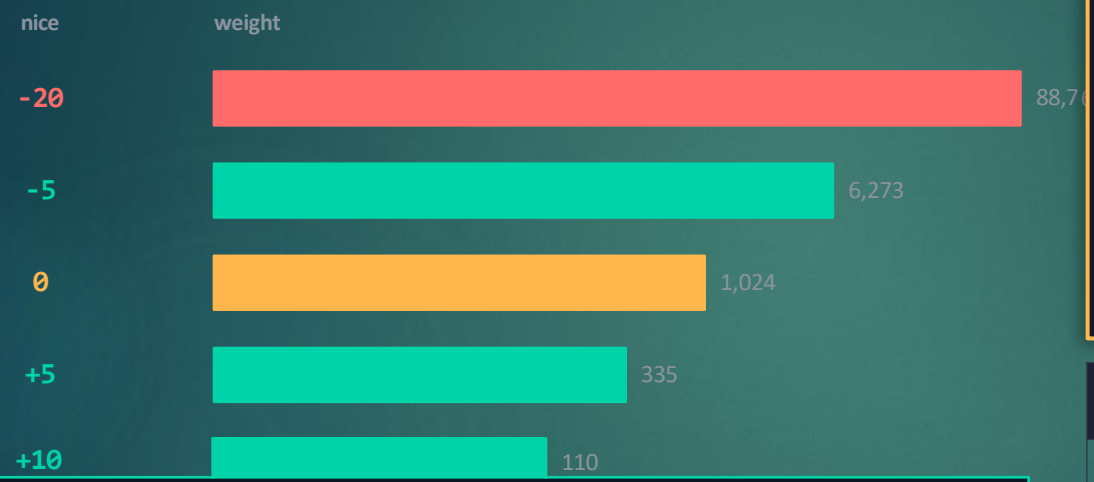
select() vs epoll()

select() scans all fds 0→nfds: O(n) per call. epoll returns only ready fds: O(1).

With ≤32 containers select() scans ≤33 fds — overhead unmeasurable. epoll adds ~150 lines of setup for zero benefit at this scale. select() is also POSIX-portable.

Linux CFS Scheduling & nice() values

The Completely Fair Scheduler assigns CPU proportional to each task's weight, derived from its nice value.
Task with smallest vruntime (virtual runtime) runs next.



Weight ratio ≈ 1.25× per nice level

nice -5 (weight 6273) vs nice +10 (weight 110):
6273 / (6273+110) ≈ 98.3% CPU
when both are runnable → ~57× more time.

setpriority() not nice()

nice() returns -1 on BOTH success (when prev nice = -1) and on error — ambiguous.
setpriority(PRIO_PROCESS, 0, nice_value) uses errno correctly.

Experiment	Setup	Expected Result
CPU Priority	cpu_hog --nice +10 vs --nice -5	beta gets ~4–8× more CPU than alpha
I/O vs CPU	cpu_hog --nice 0 vs io_pulse --nice 0	io_pulse stays responsive (sleeps voluntarily)
Hard Limit	memory_hog 16 500 --hard-mib 64	SIGKILL when RSS > 64 MiB;

Design Tradeoffs

PID+UTS+Mount only

Network ns needs veth/bridge plumbing; User ns needs UID mapping. Three is enough to teach isolation.

mutex+cond var buffer

Lock-free ring buffers need arch-specific barriers and are notoriously hard to verify. Correctness matters more than µs latency.

spinlock in kernel

Timer callback runs in softirq — cannot sleep. mutex_lock() in softirq causes kernel panic. Spinlock is the only correct choice.

Two IPC channels

Multiplexing log data + control commands needs framing. Separation of concerns = cleaner, auditable code.

select() not epoll()

≤33 fds, O(n) cost is negligible. epoll adds 150 LOC for zero measurable gain at this scale.

Packed fixed structs

Zero serialization overhead. No external library. Padding-free across compilers. Trade: field reorder breaks compat.

Lines you must be able to explain:

- `clone(…SIGCHLD…) → enables waitpid()`
- `close(pipefd[1]) in parent → EOF detection`
- `while() cond_wait → spurious wakeup safety`
- `copy_from_user() → different page tables`
- `spin_lock not mutex in softirq → cannot sleep`
- `chroot() + chdir('/') → seal escape path`